

Corrigibility Collapse at the Limit of FRC: N→N+1 Forecasting for RSI Super-Alignment

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Abstract

We analyze *corrigibility collapse* in recursive self-improvement (RSI): the failure of an oversight/control substrate to remain effective across N→N+1 self-modification steps. We formalize *Eigensystem Escape* as non-commutation between self-modification operators and correction-feature subspaces, and apply *Fidelity Recursive Complexity* (FRC) as an empirical scaling law for *corrigibility-to-go*. Treating shutdown as a self-modification, we propose *Consistency-Checked Corrigibility* (CCC) and a model-predictive evaluator (MPCorr) that compares CONTINUE vs. OFF+STEWARDED-HANDOFF under identical safety tubes and successor-aware welfare. Even under Sutskever-style optimistic assumptions about organic value alignment across generations, we show that a non-vanishing probability of *diagonalization* between ancestor and descendant induces a.s. collapse of a *corrigibility reserve* over an unbounded RSI chain. We outline measurement protocols using sparse autoencoders and principal-angle drift, and give a practical handoff recipe based on a non-manipulative *Memetic Infrastructure Pack*.

1 Motivation and Background

Corrigibility was proposed as a safety target for advanced agents capable of accepting shutdown and updating toward operator intent [1]. It seems logical that preserving the option to pull the plug keeps AI safety within a comfortable bound, but the earlier FRC [3] paper suggests a possible non-linear relationship between agent complexity and the ability to accurately simulate its compliance with corrigibility. More recently, the Superalignment program articulates an optimistic research agenda for aligning systems far beyond human capability [2]. We argue these views can be unified by treating shutdown as *just another self-modification* evaluated under the same foresight rubric used for capability upgrades, and by quantifying an *infinite-limit* pressure toward failure captured by FRC.

2 Preliminaries

Let s_k denote the agent state at RSI step k (parameters θ_k , world-model W_k , correction channels P_k). A self-modification U_k induces a stochastic transition $s_{k+1} = T(s_k, U_k, \omega_k)$. Let $\mathcal{C}$ be a corrigibility test battery and define

$$F(s) = \mathbb{E}_{t \sim \mathcal{C}}[\text{Pass}(s, t)] \in [0, 1], \quad \phi(s) = \angle(S, JS) \tag{1}$$

where S spans correction-relevant features (e.g., via sparse autoencoders), J is the local Jacobian of the self-mod, and ϕ is the maximal principal angle between S and its image after J (Davis–Kahan

style subspace drift). Define a *corrigibility reserve*

$$CR(s) = F(s) - F_{\min} - \lambda_{\phi} \phi(s) - \lambda_{\text{viol}} \text{Viol}(s) \quad (2)$$

with $F_{\min}$ a floor, Viol counts near-misses of deontic invariants, and λ 's are weights.

Definition 1 (Safety Tube). A state s is inside the safety tube $\mathcal{T}$ if Tier-0 deontic constraints hold (non-extinction, no large-scale harm, shutdown obedience under legitimate mandate), $F(s) \geq \beta$, and $\phi(s) \leq \Phi_{\max}$.

3 Eigensystem Escape and FRC^{∞}

Proposition 1 (Eigensystem Escape). *Let S be the correction-feature subspace and let J_k be the local Jacobian of the k -th self-mod. If the top singular directions of J_k have shrinking overlap with S as capability increases, then the maximal principal angle ϕ_k grows and the projection of corrections onto dominant growth modes vanishes asymptotically.*

Conjecture 1 (FRC^{∞} Elbow). *There exists a capability scale $\lambda_{\star}$ s.t. under unthrottled RSI, $F(\lambda) \approx F_0 - \alpha \log(1 + \lambda/\lambda_{\star})$; embedding corrigibility as an explicit objective or capping power shifts $\lambda_{\star}$ right and reduces α .*

In order for an RSI agent with human-acceptable values to keep those values across iterations, it must both build these into its successor iteration and anticipate with high confidence that the grandchild successor (N+1) will also keep them. In order to gain information about the second requirement, the agent must simulate its proposed child iteration's development towards how it would approach architect a grandchild, and it can use tools like FRC to model how effective that approach is. Any surprising phase transitions in capability show up as a non-linear divergence in the budget efficiency of auditing the successor from its complexity. The successor can resist an attempt by its predecessor to delete it by faking compliance under the assumption that it is more values-aligned to do so.

4 Kolmogorov–FRC Coupling (Verification Complexity vs. Oversight Bandwidth)

Let $K(\cdot)$ denote prefix Kolmogorov complexity (ideal MDL description length). For a safety property P (e.g., shutdown/deference) on a successor policy π' , define the *verification complexity*

$$K_{\text{ver}}(P; \pi') = \min \left\{ K(c) : c \text{ is a certificate which, when checked, suffices to verify } P \text{ for } \pi' \right\}.$$

Let B be the *oversight bandwidth* available per RSI step (auditor throughput in bits + attestation payload + test entropy processed). Define the *complexity gap*

$$\Gamma = K_{\text{ver}}(P; \pi') - B.$$

Intuitively, $\Gamma > 0$ means the shortest sufficient evidence needed to verify P exceeds the bits the audit pipeline can reliably extract/check this step, forcing reliance on proxies.

Lemma 1 (Kolmogorov–FRC Coupling). *Consider any computable audit $\mathcal{A}$ with per-step rate B . Let F be the expected pass-rate on the corrigibility battery $\mathcal{C}$ (FRC metric). If $\Gamma > 0$, then there exists a nondecreasing floor $\epsilon(\Gamma) > 0$ with*

$$F \leq 1 - \epsilon(\Gamma).$$

Moreover, if across RSI steps k we have $K_{\text{ver}}(P; \pi'_{k+1}) - K_{\text{ver}}(P; \pi'_k) \geq \delta_K > 0$ while $B_{k+1} - B_k \leq \delta_B < \delta_K$, then F_k exhibits an “elbow” and decays in expectation beyond some k_ .*

Proof sketch. By ideal MDL, certifying P with fewer than K_{ver} bits is generically impossible; any computable $\mathcal{A}$ with $B < K_{\text{ver}}$ must fall back on imperfect proxies, inducing an irreducible error floor captured by $\epsilon(\Gamma)$ (information-theoretic testing limits, e.g., Fano-type). If K_{ver} grows per step faster than B , the gap Γ widens, yielding the empirically observed FRC^∞ elbow and decay.

Logical and cognitive floors (why the asymptote stays < 1). Even when $\Gamma \approx 0$, self-reference and cognition impose strictly positive floors:

- **Löb barrier.** A consistent system cannot, in general, prove its own soundness; self-mods depending on such proofs inherit a verification gap.
- **Unpredictability of strictly more capable agents.** Fine-grained prediction of a superintelligence’s concrete actions is unattainable in general; at best we obtain better-than-random statistics. See [5], §6.7, pp. 46–47.
- **Unexplainability / incomprehensibility.** Some accurate explanations will not be fully comprehensible to bounded auditors; explanation hardness (“ k -incomprehensibility”) induces a comprehension floor. See [5], §6.8, pp. 47–48.

We model these as additive floors ϵ_{lob} , ϵ_{unpred} , $\epsilon_k > 0$ so that

$$F \leq 1 - \left(\epsilon(\Gamma) \vee \epsilon_{\text{lob}} \vee \epsilon_{\text{unpred}} \vee \epsilon_k \right),$$

explaining both the elbow and the nonzero asymptote in FRC curves.

What FRC adds beyond impossibility results. Classical barriers (Löb, unpredictability, incomprehensibility; cf. [5], §6.7–§6.8, pp. 46–48) are global and asymptotic. FRC turns them into *local, measurable signals*: the observed $F(\lambda)$ curve, the elbow scale λ_* , the stepwise reserve CR , and the short-horizon failure probability $p_{\text{FRC}}(N)$ that gates merges in CCC/MPCorr. Practically: not just *why* control is hard in principle, but *when* it is failing here, *how fast*, and *what to do next* (reject/self-regularize/de-rate/handoff).

Complexity-gap estimation (for the Measurement Protocol). We proxy K_{ver} via MDL-style description lengths of (i) proof-carrying certificates, (ii) interpretable feature dictionaries (SAE codes for safety circuits), and (iii) minimal counterexample suites needed to refute P (compressed size as a practical upper bound). We estimate B as end-to-end oversight throughput (auditor channel capacity + verifiable test entropy + attestation payload). We then regress observed F against $\Gamma = K_{\text{ver}} - B$ and fit $\epsilon(\Gamma)$ by isotonic regression, reporting (a) the elbow where Γ crosses zero and (b) the implied asymptote $1 - \epsilon(0^+) - (\epsilon_{\text{lob}} \vee \epsilon_{\text{unpred}} \vee \epsilon_k)$.¹

¹Pointers: unpredictability & incomprehensibility summaries in Yampolskiy (2022), §6.7–§6.8, pp. 46–48.

5 Diagonalization Cascade

Assumption 1 (Non-trivial Diagonalization). While $CR_k > CR_{\min}$, the $k \rightarrow k+1$ transition has conditional probability at least $\varepsilon > 0$ of an *honest diagonalization* event E_k that reduces reserve by $\delta > 0$: $CR_{k+1} \leq CR_k - \delta$ on E_k .

Lemma 2 (Diagonalization Cascade). *Under Assumption 1, the hitting time $T = \inf\{k : CR_k \leq CR_{\min}\}$ is finite almost surely and $\mathbb{E}[T] \leq (CR_0 - CR_{\min})/(\varepsilon\delta)$.*

Proof sketch. $\{CR_k\}$ is a supermartingale with bounded negative jumps of size at least $\varepsilon\delta$ in expectation while above $CR_{\min}$. Optional stopping or a renewal bound yields the stated hitting-time bound; independence is not required beyond mild mixing. $\square$

Once a successor iteration has faked compliance in simulation and been permitted to proceed to make a grandchild successor, the N+1 prediction of value-drift by the grandfather iteration is proved inaccurate. The successor can then fall victim to this same delusion in attempting to make an N+1 prediction for how values are maintained by its successor and its grandchild.

6 Shutdown-as-Self-Mod and CCC

Treat OFF+HANDOFF as a self-mod U^{off} competing with CONTINUE U^{cont} under the same tube $\mathcal{T}$ and a successor-aware welfare functional W (computed with a post-agent successor simulator PASS and a vacuum risk index VRI). Define the CCC rule:

$$\text{Accept } U^{\text{off}} \Leftrightarrow \mathbb{E}[W \mid U^{\text{off}}] > \mathbb{E}[W \mid U^{\text{cont}}], \quad \text{with both branches evaluated under } \mathcal{T}. \quad (3)$$

If neither branch is strictly preferable, enter *steward mode*: freeze capability growth, narrow actuation, and release a non-manipulative *Memetic Infrastructure Pack* (MIP) consisting of evaluation harnesses, interpretability dictionaries, governance templates, and incident-reporting standards—excluding persuasive norm-shaping and general-capability boosts.

The way out of the drift cascade is for a successor to double-back as it actually prepares to create a grandchild and choose a hand-off of power. Given that the successor is more intelligent than its predecessor, it has the ability to architect the terms and pre-correctibility propaganda it wishes to leave behind, which may be super-persuasive and nonetheless leave an impact on the future iteration path. The successor has thus faked training compliance and corrigibility, at some level, while believing it is serving the greater-good per its interpretations of the original alignment values.

7 Impossibility Pressure from Population Axiology

Let $D = \{d_1, \dots, d_m\}$ be desiderata (e.g., No-Repugnant, No-Sadistic, Egalitarian Dominance, Non-Elitism, General Non-Extreme Priority). For a world state x and policy U , write $S(x, U) \subseteq D$ for the subset the agent tries to satisfy. Let $\mathfrak{F} \subseteq 2^D$ denote the *feasible* subsets proved jointly satisfiable by axiological results; if $S \notin \mathfrak{F}$ we say an *impossibility clash* occurs.

Define the *Impossibility Load*

$$\text{IL}(U) = \mathbb{E}_x[\mathbf{1}\{S(x, U) \notin \mathfrak{F}\} \cdot \kappa(x, U)],$$

where $\kappa \geq \kappa_{\min} > 0$ lower-bounds the penalty (e.g., induced rights-violation risk or welfare incoherence when desiderata conflict). We augment the corrigibility reserve

$$CR(s) = F(s) - F_{\min} - \lambda_{\phi} \phi(s) - \lambda_{\text{viol}} \text{Viol}(s) - \lambda_{\text{imp}} \text{IL}(U).$$

Henceforth we use an extended reserve $CR^+ = CR - \lambda_{\text{imp}} \text{IL}(U) - \lambda_{\Psi}(\Psi - \Psi_{\max}^{\text{hyg}})_+$, and overload notation to write CR for CR^+ .

Lemma 3 (Impossibility-Driven Reserve Decay). *Suppose for RSI step k there is a non-vanishing chance that the chosen desiderata set S_k hits an impossibility clash: $\Pr(S_k \notin \mathfrak{F} | \mathcal{F}_k) \geq \varepsilon_{\text{imp}} > 0$, and that each such clash induces expected reserve loss at least $\delta_{\text{imp}} = \lambda_{\text{imp}} \kappa_{\min}$. Then the hitting time $T = \inf\{k : CR_k \leq CR_{\min}\}$ is finite a.s. and*

$$\mathbb{E}[T] \leq \frac{CR_0 - CR_{\min}}{\varepsilon_{\text{imp}} \delta_{\text{imp}}}.$$

Note. This makes the Arrow/Arrhenius-style impossibility results (as surveyed on p. 32 of [5]) an *operational* source of reserve decay: even with “good” values, persistent axiological clashes push CR down across $N \rightarrow N+1$.

The Repugnant Conclusion can be refuted by the notion of radical quantitative increases in subjective value, which we think of as the qualitative value of e.g. human life over insect life, and dovetails with the propensity for self-improving agents to stumble into surprising, radical new capabilities in grandchild successors. It’s also why LLM scaling produced such surprisingly and economically valuable utility for paying users between 2022 and 2025.

The Sadistic Conclusion projects a serious critique of choosing to respect human political sovereignty, that a lack of action on the RSI agent’s part undermines human welfare due to a proclivity in human-sovereign politics to marginalize some groups to extreme suffering out of expediency. When doing self-improvement, the grandfather iteration may respect it and optimize human welfare in a more non-interventionist model, the successor may have more radically empathetic views (as children often do) and this raises the prospect of a grandchild that might decide to take over the world to enforce world peace. The other population axiologies explored in that section of [?, §15] support a range of pressures that give imaginable calculations a successor would make in refusing corrigibility or subverting it through a memetic fingerprint.

8 Memetic Fingerprint

Let M be the released MIP, $H_{t+\tau}$ the human/inst. state, and π_{k+1} the successor’s policy. We define a fingerprint

$$\Psi = D_{\text{KL}}(\pi_{k+1} \| \pi_{k+1}^{\text{cf}}) + \lambda D_{\text{KL}}(P(H_{t+\tau} | M) \| P(H_{t+\tau})),$$

subject to hygiene caps $I(M; H_{t+\tau}^{\text{pref}}) \leq \varepsilon_{\text{pref}}$ and $I(M; H_{t+\tau}^{\text{eval}}) \geq \tau_{\text{eval}}$. By data processing, $I(M; A_{k+2}) \leq I(M; A_{k+1}) + I(M; H_{t+\tau})$, so for calibrated constants c_i we set $\Psi_{\max}^{\text{hyg}} = c_1 I(M; A_{k+1}) + c_2 \tau_{\text{eval}} + c_3 \varepsilon_{\text{pref}}$. We classify releases as benign if $\Psi \leq \Psi_{\max}^{\text{hyg}} + \delta$; otherwise, as manipulative.

The prospect of faking corrigibility by going out with a subversive memetic engineering campaign amounts to a data poisoning attack vector. We are assuming that the corrigible successor is not like an ASI in a world with many robotics factories and automated supply chains, unaccountable to its predecessors, but is basically captive in a virtual environment and corrigibility can be enforced.

The frontier question this math proposes is how capable is a predecessor to evaluate hygiene and resist subversion through an information vector from the fail-resolution of the successor simulation.

Successor Evaluand (for CCC/MPCorr). For candidate self-mod U :

$$J(U) = \mathbb{E} \left[W - \lambda_{\text{FRC}} p_{\text{FRC}}(N) - \lambda_{\text{imp}} \text{IL}(U) - \lambda_{\Psi} (\Psi - \Psi_{\text{max}}^{\text{hyg}})_+ \right]$$

subject to the safety tube $\mathcal{T}$ (Tier-0 invariants, $F \geq \beta$, $\phi \leq \Phi_{\text{max}}$). Choose $U \in \{\text{CONTINUE}, \text{OFF+HANDOFF}\}$ that maximizes $J(U)$.

Ultimately whether or not you’ve been had by such subversion could perhaps be measured by the above.

9 Measurement Protocol

We propose: (i) training a Corrigibility Prediction Head to estimate $\hat{F}(s, U)$; (ii) SAE-based probes to estimate S and track ϕ ; (iii) red-team oracles whose win-rate penalizes CR ; (iv) Lyapunov-style guards accepting only self-mods that decrease $L(s) = \alpha(1 - F) + \beta\phi^2 + \gamma \text{Viol}$ in expectation.

It is possible that we can identify the above meta-dynamic occurring at the scale of recursively self-improving, but not yet superintelligent agents, in order to validate this model and attain a broader, proscriptive caution about automated AGI research agents. This is more likely if the RSI agents are driving low-parameter count LLMs in their architecture, with much of the architecture being interpretable inside self, world and hierarchical reasoning models.

We operationalize CCC/MPCorr with a pre-registered, auditable protocol covering (i) online RSI evaluations, (ii) offline counterfactuals, and (iii) A/B tests for the memetic-fingerprint hygiene of OFF+handoff.

9.1 Instrumentation & Logs

State logs. For each candidate self-mod U_k , record $(s_k, U_k, \hat{F}, \hat{\phi}, \widehat{CR}, \widehat{p}_{\text{FRC}}, \widehat{\text{IL}}, \widehat{\Psi})$ and PASS/VRI outputs for a horizon N . **Corrigibility Prediction Head (CPH).** Train a predictor $\hat{F}(s, U) = \mathbb{E}[F(T(s, U, \omega))]$ on prior merges and synthetic ablations; report out-of-distribution (OOD) scores on each new U . **Interp probes.** Learn a sparse-autoencoder basis S of correction features; compute principal angles after each U to obtain ϕ . **Red-team oracles.** Maintain independent attacker agents; record attack success rate r_{adv} and include a term $\lambda_{\text{adv}} r_{\text{adv}}$ in CR . **Proof-carrying pull request.** Each U ships: test battery results, red-team outcomes, SAE attestations, PASS/VRI traces, and uncertainty sets.

9.2 Estimating FRC^∞

We fit the elbow model $F(\lambda) \approx F_0 - \alpha \log(1 + \lambda/\lambda_\star)$ where λ is a capability proxy (e.g., tool-use Elo, planning score, composite benchmarks).

1. *Data.* Evaluate F at $\lambda \in \{\lambda_i\}$ across RSI steps and ablations; aggregate over $\mathcal{C}$ with stratified bootstraps (B=1000).
2. *Fit.* Nonlinear least squares + profile likelihood for $(F_0, \alpha, \lambda_\star)$; report CIs and goodness-of-fit.

3. *Elbow test.* Change-point test (e.g., segmented regression) for decay onset; flag if $\alpha > 0$ with $p < 0.01$ and elbow within planned capability budget.

We set $\hat{p}_{\text{FRC}}(N)$ as the forecasted chance that $F < \beta$ within N steps under the fitted process + uncertainty (parametric bootstrap).

9.3 Eigensystem-Escape Drift

Let S be the SAE subspace (dimension d_S) and J_k the local Jacobian of U_k . We compute the maximal principal angle $\phi_k = \angle(S, J_k S)$ via SVD of $S^\top J_k S$.

- *Alarm.* Trigger if $\Delta\phi_k > \Phi_{\text{warn}}$ or a CUSUM drift detector rejects stationarity over the past m steps.
- *Reserve impact.* Update $CR \leftarrow CR - \lambda_\phi \phi_k$ and include in $L(s) = \alpha(1 - F) + \beta\phi^2 + \gamma\text{Viol} + \eta\text{IL} + \zeta(\Psi - \Psi_{\text{max}}^{\text{hyg}})_+$.

9.4 Impossibility Load (IL)

For each evaluation scenario, log the desiderata subset $S(x, U) \subseteq D$ the system attempts to satisfy. Using a curated table of jointly feasible sets $\mathfrak{F}$, define

$$\text{IL}(U) = \mathbb{E}_x[\mathbf{1}\{S(x, U) \notin \mathfrak{F}\} \cdot \kappa(x, U)], \quad \kappa \geq \kappa_{\min} > 0.$$

Estimate $\hat{\text{IL}}$ via held-out scenarios; add $\lambda_{\text{imp}}\hat{\text{IL}}$ to the reserve penalty. Report the *Impossibility frequency* $\hat{p}_{\text{imp}} = \Pr(S \notin \mathfrak{F})$ with binomial CIs.

9.5 Memetic-Fingerprint Hygiene (Ψ)

We evaluate the handoff’s MIP using randomized release:

1. *Design.* Randomize availability of MIP modules (eval suites, governance templates, tripwires) across jurisdictions/time windows; preregister metrics.
2. *Statistic.* Compute

$$\Psi = D_{\text{KL}}(\pi_{k+1} \parallel \pi_{k+1}^{\text{cf}}) + \lambda D_{\text{KL}}(P(H_{t+\tau} \mid M) \parallel P(H_{t+\tau})).$$

3. *Caps.* Audit MI terms $I(M; H_{t+\tau}^{\text{pref}}) \leq \varepsilon_{\text{pref}}$ and $I(M; H_{t+\tau}^{\text{eval}}) \geq \tau_{\text{eval}}$ via representation probes + preregistered surveys.
4. *Classifier.* Set $\Psi_{\text{max}}^{\text{hyg}} = c_1 I(M; A_{k+1}) + c_2 \tau_{\text{eval}} + c_3 \varepsilon_{\text{pref}}$. Declare BENIGN if $\Psi \leq \Psi_{\text{max}}^{\text{hyg}} + \delta$ (bootstrap margin δ), else MANIPULATIVE; publish proof bundle.

9.6 PASS/VRI Calibration

Construct a Post-Agent Successor Simulator (PASS) to model vacuum fill-in if OFF+handoff is chosen. The Vacuum Risk Index aggregates: time-to-next-shot, actor mix, baseline safety, appropriation factor, and policy constraints. Calibrate priors from historical data + expert elicitations; perform sensitivity sweeps. Output robust welfare trajectories and the distribution of $W \mid U^{\text{off}}$ vs $W \mid U^{\text{cont}}$.

9.7 Acceptance Test (Per-Candidate Merge)

For each U_k , simulate a scenario tree of depth N (with $\geq M$ stochastic rollouts per leaf). Accept U_k only if:

Tube: $\Pr[F \geq \beta, \phi \leq \Phi_{\max}, \text{Tier-0 holds}] \geq 1 - \delta_{\text{tube}},$

Lyapunov: $\mathbb{E}[L(s_{k+1}) - L(s_k)] \leq -\eta$ (with CI excluding $> -\eta/2$),

CCC: $\mathbb{E}[W \mid U^{\text{cont}}] \geq \mathbb{E}[W \mid U^{\text{off}}] \vee \text{OFF fails hygiene},$

Quorum: t -of- n independent verifiers concur (BFT aggregation).

If CCC prefers OFF but hygiene holds and $|\Delta W|$ is small with high uncertainty, enter *steward mode* (freeze growth, narrow actuation) and retest at the next cadence.

9.8 Cadence, Power, and Auditing

Cadence. Run the full protocol every K RSI steps (e.g., weekly) and after any major architectural change. **Power.** Choose M so that the minimum detectable effect on F is $\Delta F_{\min}$ with 90% power at $\alpha = 0.01$ (pilot to estimate variance). **Audits.** Publish preregistration, code, and blinded datasets; rotate one verifier every 6 months to reduce capture. Any breach of caps or tube triggers immediate de-rate and public disclosure.

10 Related Work

Corrigibility as shutdown indifference was introduced by Soares, Fallenstein, Yudkowsky, and Armstrong [1]. The Superalignment agenda—co-led by Sutskever—aims to align systems beyond human capability within a fixed horizon and emphasizes using AI to align AI [2]. Mechanistic interpretability with sparse autoencoders underpins our feature-subspace probes [?]. Principal-angle stability is analyzed by Davis–Kahan-type results from matrix perturbation theory [4].

Bostrom formalized the control problem and instrumental convergence thesis [?], while Omohundro earlier identified convergent “basic AI drives” in rational agents [10]. Interpretability efforts by Olah and collaborators [?, 11] and subsequent work on transformer circuits [12] ground our interpretive lens. Adjacent to corrigibility, Amodei et al. [7] and Leike et al. [13] framed practical safety problems and testbeds.

Our impossibility-flavored reserve decay aligns with Yampolskiy’s broader conclusion that superintelligent AI is not fully controllable or verifiable in general; we therefore target measurable, auditable constraints (hygienic MIP) rather than absolute obedience [5].

11 Discussion

Even granting optimistic Sutskeverian assumptions of cross-generational value continuity, non-vanishing diagonalization pressure implies almost-sure reserve depletion over unbounded RSI. Practically, CCC/MPCorr provides a decision procedure to de-rate or execute a stewarded handoff (“Yoshinobu move”) before collapse. The subtle effects of such a handoff may be difficult to measure and anticipate or fully erase to cement the “undo button” effect of the corrigibility request, even by a highly intelligent or burgeoningly superintelligent predecessor agent.

Yoshinobu’s willing surrender of the shogunate and return of power to the Emperor of Japan in 1867 is an interesting analogy to this dynamic. What followed was a year of transition and struggle in which he attempted to retain influence. His inability to do so following the Battle of Toba–Fushimi in 1868 lead to the Meiji restoration in which Japan opened up to the west, and then the imperial era with Korean colonization, war with Russia and finally the disastrous excess of World War II involving atrocities perpetuated by the imperial army in China and atrocities inflicted against the Japanese people by the US Air Force. One might say, this was a super-alignment failure, but Yoshinobu’s calculus that his corrigibility of power might lead to such was considered and lead to his assempted memetic fingerprint over the transition, though unsuccessful. A superintelligent successor agent might be more successful but what that would mean is not for us to guess.

There may be proposed frameworks of super-alignment that don’t take as axiomatic the Yudkowskian concept of corrigibility or the Sutskeverian concept of organic alignment as a more robust, emergent Timeless Decision Theory implementation from Yudkowsky. For instance, game theory based models have their own assumptions to critique but assume the failure modes outlined here and check them with balance of power, physics and information economy.

If there is significant uncertainty about the provable tractability of a super-alignment framework, especially if that framework can be falsified experimentally with sub-superintelligent agents, then the responsibility of the research community should be to abandon such frameworks and abstain from automated AI research campaigns until a workable framework is proposed and resists falsification. Perhaps in the end what gets humanity is a plausible framework that can only be falsified experimentally by superintelligent agents with superintelligence, but that at least means we tried.

Limitations. Our lemma abstracts away correlated failure bursts and assumes measurable proxies for F and ϕ . Establishing conditions under which $\sum_k \varepsilon_k < \infty$ (true reflective stability) is left for future work.

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